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PY211 Spring 2026 Homework 2 **due date 11:59 pm Saturday Jan. 31st**

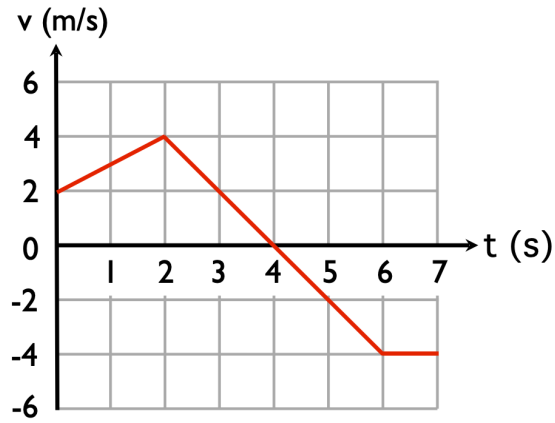
Complete The Expert TA Assignment 2, and answer these additional two pages of problems. For these problems show your work, and upload to Gradescope by the deadline. You must use our template. Print this out, or download it, fill in your answers in the space provided, and then upload your work to Gradescope as a 2-page PDF. The pages must be in the correct order.

PROBLEM 1 – show work in the space provided

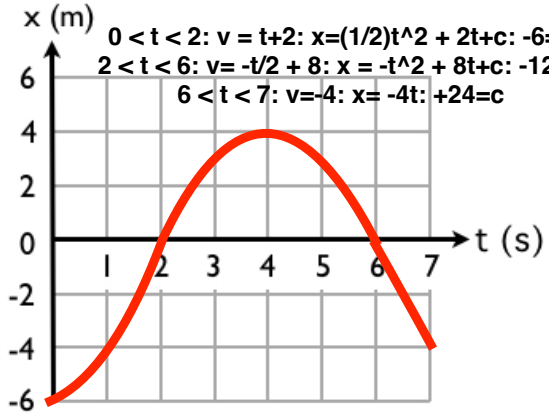
The graph shows the velocity of an object as a function of time between t = 0 s and t = 7 s.

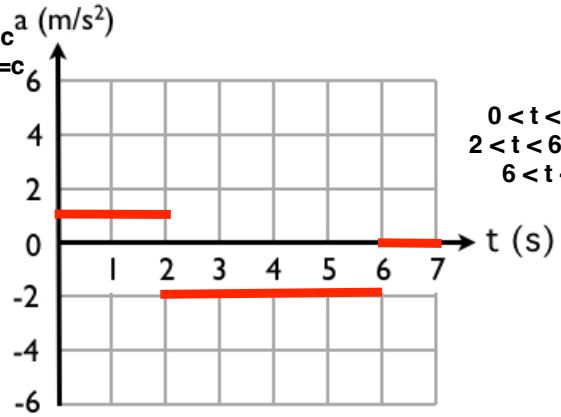
The object is just moving in one dimension (along a straight line).

[3 points] (a) On the axes below, plot the position of the object as a function of time, and the object's acceleration as a function of time. Note that the object is located at x = -6.0 m at t = 0 s.



0 < t < 2: v = t+2: x=(1/2)t^2 + 2t+c: -6=c
2 < t < 6: v= -t/2 + 8: x = -t^2 + 8t+c: -12=c
6 < t < 7: v=-4: x= -4t: +24=c





0 < t < 2: v = t+2: a=1
2 < t < 6: v= -2t + 8: a=-2
6 < t < 7: v=-4: a=0

[1 point] (b) What is the total distance traveled during the interval from t = 0 s to t = 7 s?

6 + (8*4)/2 + 4 = 26m

distance = 26m

[1 point] (c) What is the average velocity during the interval from t = 0 s to t = 7 s?

\int_0^7{a dt} = 2 + (4*-2) + 0 = -6

$\overrightarrow{v_{av}}$ = -6 m/s

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PROBLEM 2 – show work in the space provided

The position (as a function of time) of an object that is moving along the x-axis is given by:

$$x(t) = +20\ m + (3.0\ m/s)t - (1.0\ m/s^3)t^3$$

[1 point] (a) What is the expression for the object's velocity as a function of time?

v(t) = **3–3t^2**

[1 point] (b) What is the expression for the object's acceleration as a function of time?

a(t) = **-6t**

[1 point] (c) What is the object's average velocity between *t* = 0 s and *t* = 4 s?

vavg = (x(4)-x(0))/4 = (-32-20)/4 =-13

$\vec{v}_{av} =$ **-13m/s**

[1 point] (d) What is the total distance traveled by the object between *t* = 0 s and *t* = 4 s?

Δx = **56m**

d=|x(1)–x(0)|+|x(4)–x(1)| = 2 + 54 = 56

Check the box for a free point if you have used our template for your assignment and uploaded a 2-page PDF with the pages in the correct order.